## Aim of the Experiment :-

To determine the overall thermal conductivity of insulating material, measure thermal conductivity using a guarded hot plate or heat flow meter.

## Theory :-

A composite wall comprises a number of layers of different materials. widths and thermal conductivity. composite walls find in many applications viz in furnaces. electrical cables, nuclear reactors, pipe lines carrying steam etc. The heat conducted through composite wall is given by.

$$\dot{Q} = \frac{\Delta T}{\Sigma R_{total}}$$

$$\Sigma R = R_{mild\,steel} + R_{Hylam} + R_{wood}$$

where,

$\Delta T$ = overall temperature difference in °C.

R = Thermal resistance of plate can be calculated by

$$R_{th} = \frac{L}{K}$$

where,

L = length of the plate in m.

K = Thermal conductivity of material in (W/mK)

$$K = \frac{R_{th}}{L}$$

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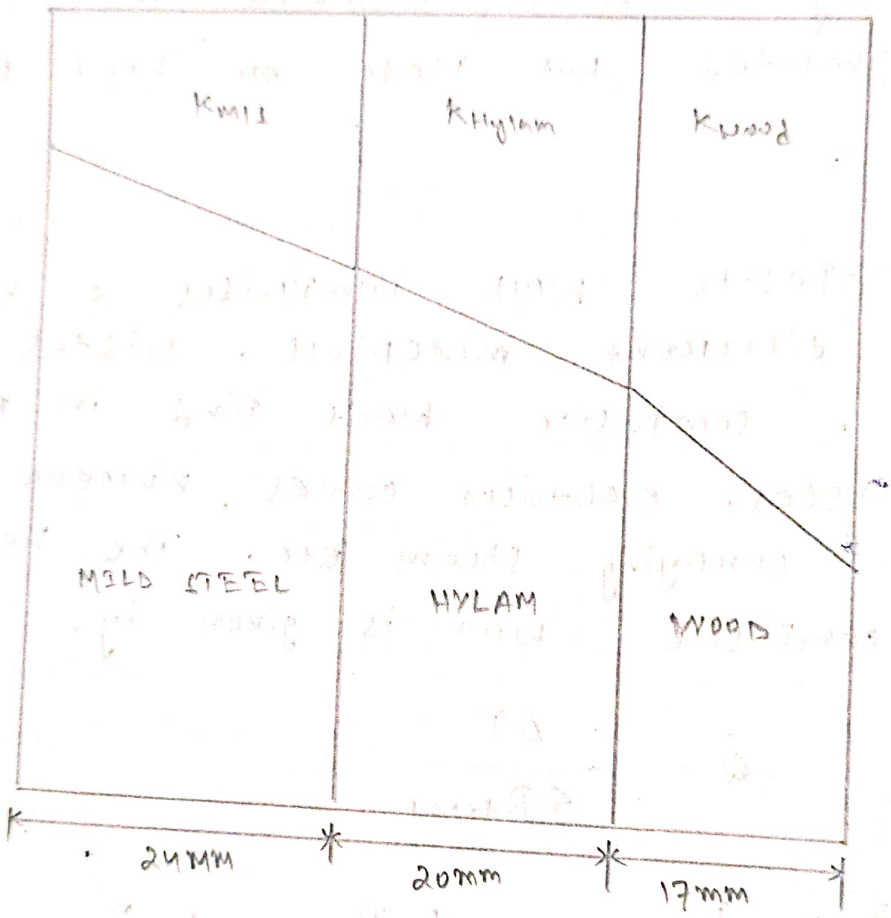


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Specification :-

-> Heat coil type, cap, 250W
-> Heat control unit (variac) = 0-230V, 0-2 Amp
-> voltmeter Range :- 0-300V
-> Ammeter Range :- 0-20 Amp
-> Temperature indicator (Digital type) Range : 0-800°C
power required - A.C single phase 230V.
-> Thermocouple used : Teflon coated, chromal-Alumal.
-> Thickness of slabs M.S - 24 mm
Hylam - 20 mm
Wood - 17 mm
-> Diameter of the each slab, d = 150 mm = 0.15m
-> Thermal conductivity of steel $K_1$ = 45 W/m.K
-> Thermal conductivity of Hylam $K_2$ = 4 W/m.K
-> Thermal conductivity of wood $K_3$ = 0.12 - 0.04 W/mK

Description of Apparatus :-

The apparatus consist of three slabs made of mild steel, Hylam & wood having same thickness. The three slabs are clamped on both sides using bolts and nuts. on one side of the composite wall, a heater is provided. The heater input can be varied using in a dimmer stat thermocouples are embedded at three different places in each section to find out the average temperature.

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on the outside of the wall, cooling water jacket is provided which takes away the heat conducted through the slabs. The mass flow rate of water and its raise in temperature can determined.

Procedure :-

-> The equipment had been connected to the power supply.

-> The power input had been adjusted to the required value.

-> Sufficient time had been allowed for the system to attain a steady state.

-> All temperatures had been noted down by operating the knob.

-> The experiment had been repeated for different heat inputs.

Observation :-

Heat Input :-

-> Volt meter reading V : 60 volt

-> Ammeter reading I : 0.53 Amp

-> Heat input V × I : 31.8 watt.

Heat flow per side $= \frac{V \times I}{2} = \frac{31.8}{2} = 15.9$ watt.



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Tabulation :-

| Sl No. | Heat I/P V (Volt) | Heat I/P I (Amp) | V×I (W) | $T_1$ | $T_2$ | $T_3$ | $T_4$ | $T_5$ | $T_6$ | $T_7$ | $T_8$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 01 | 60 | 0.53 | 31.8 | 84 | 83 | 81 | 82 | 77 | 75 | 36 | 39 |
| 02 | 60 | 0.53 | 31.8 | 87 | 86 | 84 | 85 | 79 | 76 | 36 | 40 |
| 03 | 60 | 0.53 | 31.8 | 90 | 88 | 86 | 87 | 80 | 77 | 37 | 41 |

Calculation :-

Temperature of Hot end, $T_h = \frac{T_1 + T_2}{2} + 273.15K$

$T_h = \frac{84 + 83}{2} + 273.15$

$T_h = 356.65K$

Temperature of cold end, $T_c = \frac{T_7 + T_8}{2} + 273.15K$

$T_c = \frac{36 + 39}{2} + 273.15K$

$T_c = 310.65K$

Average Temperature of Each slab :-

→ $(T_{MS})$ mild steel $= \frac{T_3 + T_4}{2} = \frac{81 + 82}{2} + 273.15$

$(T_{MS})$ mild steel $= 354.65K$.

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→ $T_{Hylam} = \frac{T_5 + T_6}{2}$

$T_{Hylam} = \frac{77 + 75}{2} + 273.15K$

$T_{Hylam} = 349.15K$

→ $T_{wood} = \frac{T_7 + T_8}{2}$

$T_{wood} = \frac{36 + 39}{2} + 273.15K$

$T_{wood} = 310.65K$

→ To find thermal conductivity of each slab we know that

$$Q = \frac{\Delta T}{\Sigma R_{th}} = \frac{T_h - T_{m/s}}{R_{m/s}} = \frac{T_{m/s} - T_{Hylam}}{R_{Hylam}}$$

$$= \frac{T_{Hylam} - T_{wood}}{R_{wood}} = \frac{T_{wood} - T_c}{R_{wood}}$$

Thermal conductivity of mild steel

$$Q = \frac{T_h - T_{m/s}}{R_{m/s}}$$

$$R_{m/s} = \frac{T_h - T_{m/s}}{Q}$$



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$$R_{m/s} = \frac{356.65 - 354.65}{15.9}$$

$$R_{m/s} = 0.125 \ K/W$$

and, $$R_{m/s} = \frac{L_{m/s}}{K_{m/s} \times A_{m/s}}$$

$$K_{m/s} = \frac{L_{m/s}}{R_{m/s} \times A_{m/s}}$$

$$A_{m/s} = \frac{\pi}{4} \times (d)^2 = \frac{\pi}{4} \times (0.15)^2 = 0.0176 \ m^2$$

W.K.T

$$A_{m/s} = A_{Hylam} = A_{wood} = 0.0176 \ m^2$$

$$K_{m/s} = \frac{0.024}{0.125 \times 0.0176}$$

$$K_{m/s} = 10.90 \ W/mK$$

similarly the thermal conductivity of Hylam is equal to

$$Q = \frac{T_{m/s} - T_{hylam}}{R_{Hylam}}$$

$$R_{Hylam} = \frac{T_{m/s} - T_{Hylam}}{Q} = \frac{354.65 - 349.15}{15.9}$$

$$R_{Hylam} = 0.345 \ K/W$$

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$$R_{Hylam} = \frac{L_{Hylam}}{K_{Hylam} \times A_{Hylam}}$$

$$K_{Hylam} = \frac{L_{Hylam}}{R_{Hylam} \times A_{Hylam}}$$

$$K_{Hylam} = \frac{0.020}{0.345 \times 0.0176}$$

$$K_{Hylam} = 3.293 \ W/mK$$

similarly for wood the thermal conductivity will be

$$Q = \frac{T_{Hylam} - T_{wood}}{R_{wood}}$$

$$R_{wood} = \frac{327.15 - 310.65}{15.9}$$

$$R_{wood} = 1.037 \ K/W$$

$$R_{wood} = \frac{L_{wood}}{K_{wood} \times A_{wood}}$$

$$K_{wood} = \frac{0.017}{1.037 \times 0.0176}$$

$$K_{wood} = 0.931 \ W/mK.$$

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Conclusion :-

From the above experiment we found the thermal conductivity of mild steel $k_{m/s} = 10.90$ W/mK and the thermal conductivity of Hylam $k_{Hylam} = 3.293$ W/mK and lastly the thermal conductivity of wood $k_{wood} = 0.931$ W/mK Respectively.

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